Direction-Aware 3D Haptic Retargeting in VR

Zhikai Wen
Beihang University

Jian Wu
Beihang University

Runze Fan
Beihang University

Sio Kei Im
Macao Polytechnic University

Lili Wang*
Beihang University

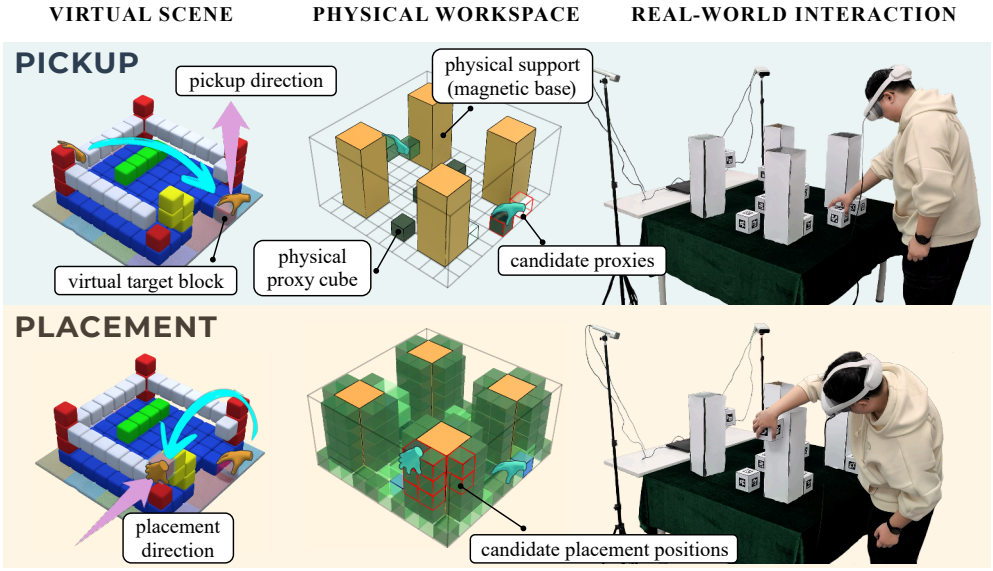


Figure 1: Overview of a user performing a 3D block-stacking task with direction-aware haptic retargeting. The top and bottom rows show pickup and placement, respectively. In the left column, blue arrows trace the hand or held-block motion and purple arrows indicate the intended interaction direction. The middle column visualizes the scored physical workspace: darker green denotes a higher Haptic-Direction Score, red wireframes mark candidates retained by Candidate Filtering, and yellow/green hands denote the virtual/real hands. The highest-scoring feasible proxy or placement position is selected. The right column shows the user interacting with the magnetic physical proxy system.

Abstract

Passive haptic retargeting effectively addresses real-world limitations in virtual reality by repurposing physical proxies. However, existing methods typically rely on single proxies, 2D planar arrangements, or fixed 3D proxies, without jointly managing reusable proxies during volumetric stacking. This limitation stems from the challenges of managing mobile proxies in volumetric space while ensuring directional consistency between virtual and physical contacts. To address these problems, this paper proposes a direction-aware 3D haptic retargeting method. We first design a magnetic physical proxy system that enables stable 3D stacking and dual-camera, marker-based pose tracking. Central to our approach is the introduction of a Haptic-Direction Score, which quantitatively evaluates non-occupied proxies and available placement positions with Local Exposure, Global Haptic Direction Quality, and Non-occupied Proxy Availability. We also propose a haptic retargeting method based on Haptic-Direction Score, which dynamically maps virtual targets to optimal physical targets during the retargeting process. A user study comparing our approach with three planar methods adapted to 3D demonstrates that our method significantly reduces the system reset rate, while improving task completion efficiency, haptic quality, and overall user experience in the evaluated cubic-stacking tasks.

Index Terms: Haptic Retargeting, Hand Redirection, Perception, Reset Techniques, Virtual Reality.

1 Introduction

With the rapid advancement of information technology, Virtual Reality (VR) has been widely applied in various fields such as industrial design, education, and entertainment. In these immersive environments, haptic feedback plays a crucial role in enhancing the sense of presence and spatial perception, as it bridges the gap between digital throughput and physical sensation. Generally, VR haptics are categorized into active haptics and passive haptics. Active haptics rely on actuated devices, such as force-feedback exoskeletons, to dynamically simulate physical resistance. While flexible, they are often limited by high hardware costs and mechanical complexity. In contrast, passive haptics [17] use physical proxies to provide physical contact and motion-constraining kinesthetic support, offering a low-latency, cost-effective solution for tangible interaction. However, providing a one-to-one physical proxy for every virtual object is often impractical due to limited physical space. To address this, researchers have proposed redirected touching and haptic retargeting techniques.

The concept of redirected touching was pioneered by Kohli et al. [19], who introduced a spatial warping mechanism that maps multiple virtual objects to a single physical proxy, providing pas-

*Corresponding author
Zhikai Wen, Jian Wu, Runze Fan and Lili Wang are with the State Key Laboratory of Virtual Reality Technology and Systems, Beihang University, Beijing, China. E-mail: wenzhikai@buaa.edu.cn, lanayawj@buaa.edu.cn, by2106131@buaa.edu.cn, and wanglily@buaa.edu.cn.
Sio Kei Im is with Macao Polytechnic University. E-mail: marcusim@mpu.edu.mo.

2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR) | 979-8-3195-2565-9/26/$31.00 ©2026 IEEE | DOI: 10.1109/ISMAR70079.2026.00077

2473-0726/26/$31.00 ©2026 IEEE
DOI 10.1109/ISMAR70079.2026.00077

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

sive haptic feedback for a range of virtual targets. Based on this, Azmandian et al. [3] formalized the haptic retargeting framework by introducing body warping and world warping, which leverage visual dominance to bridge spatial discrepancies between virtual and physical environments in 3D space. Subsequently, researchers have proposed various optimization strategies to refine proxy selection or placement to minimize the frequency of system resets and enhance the continuity of interaction [21, 22]. However, many of these approaches either rely on a single proxy or arrange multiple proxies on a 2D plane, limiting their ability to provide directionally consistent 3D haptic feedback for complex spatial tasks that involve reusable proxies and evolving load-bearing dependencies, such as volumetric stacking.

To address this, some studies have explored specialized hardware, such as hemispherical stands [6], to position physical proxies in 3D configurations, supporting physical contact at multiple spatial locations. Despite these advances, persistent 3D passive-haptic retargeting with reusable proxies still faces several challenges. One challenge is that managing the placement and selection of mobile physical proxies in 3D space is non-trivial. It requires sophisticated optimization to achieve a low reset rate and high interaction efficiency, ensuring that the physical workspace remains adaptive to dynamic virtual tasks. Another challenge lies in ensuring consistency between the physical orientation of 3D proxies and the pickup or placement direction of the virtual objects they represent, as well as the support direction of other proxies or virtual objects for the current proxy or object, which is required for directionally supported contact without geometric conflicts.

Volumetric stacking couples load-bearing dependencies, exposed approach faces, and future proxy availability, requiring joint optimization beyond planar or single-boundary retargeting.

Cube-based volumetric stacking serves as a controlled abstraction of sequential 3D assembly and construction tasks in VR, such as assembly training, assembly planning, and tangible prototyping. This task class matters because the usefulness of passive-haptic retargeting in such scenarios depends on sustaining a sequence of physically supported contacts while reusing a limited set of physical proxies, rather than providing isolated touches. Without jointly managing load-bearing dependencies and future proxy availability, a mapping that is feasible for the current action can lock a supporting proxy into the structure, block access to an approach face required later, or make a later-needed proxy unavailable, causing subsequent contact to become infeasible or directionally unsupported and forcing a reset.

We scope the Score to uniform 1:1 axis-aligned cubes, orthogonal stacks, six cardinal directions, and at most five exposed faces after attachment; the magnetic base is one implementation of stable support and is not a movable proxy.

In this paper, we propose a direction-aware 3D haptic retargeting method to address these challenges. We first design a magnetic physical proxy system consisting of magnetic bases and magnetic blocks with ArUco markers, which allows for stable 3D stacking and dual-camera, marker-based pose tracking. Then, we introduce the concept of non-occupied proxies that are both available and geometrically reachable, and their associated Haptic-Direction Score. By calculating this Haptic-Direction Score, we quantitatively evaluate each candidate proxy based on Local Exposure, Global Haptic Direction Quality, and Non-occupied Proxy Availability. We also propose a haptic retargeting method based on Haptic-Direction Score, which enables our physical proxy pick-and-place and reset strategies to dynamically map virtual targets to optimal physical locations. Finally, we design and conduct a user study to evaluate our approach. The results demonstrate that, compared to three planar methods adapted to 3D, our method significantly reduces the reset rate while improving task completion efficiency, haptic quality, and the overall user experience in the evaluated cubic-stacking tasks.

Figure 1 illustrates a complete pickup-and-placement cycle using our direction-aware 3D haptic retargeting method. It links the virtual action, the candidate scores in the physical workspace, and the resulting real interaction, making explicit how intended direction and future proxy availability affect each mapping decision.

In summary, the main contributions of our work are as follows:

- We propose a direction-aware 3D haptic retargeting method that supports efficient retargeted interaction and directionally supported physical contact in the evaluated cubic-stacking tasks.
- We design a magnetic physical proxy system, introduce the concept of non-occupied proxies and their Haptic-Direction Score, and provide a method for computing the Haptic-Direction Score.
- We propose Haptic Retargeting based on Haptic-Direction Score.
- We design a user study to evaluate the performance of our method.

2 Related Work

In this section, we briefly introduce the related prior work, including passive haptics and retargeting, perception and interaction quality, and reset reduction strategies.

2.1 Passive Haptics and Retargeting

Passive haptics use unactuated physical props to provide contact and motion-constraining reaction forces for virtual objects. Our cubes provide contact and kinesthetic support rather than active force rendering or separately validated cutaneous realism. To address the impracticality of one-to-one virtual-physical mappings, early works such as Kohli et al. [19] demonstrated the potential of reusing a single physical prop to simulate multiple virtual tools. Nilsson et al. distinguish proxy similarity from co-location [26]; here cubes fix similarity and retargeting supplies co-location. Building on this, Azmandian et al. [3] formally introduced Haptic Retargeting, leveraging visual dominance to warp the user's virtual hand or the virtual world to guide the user to a shared physical prop. Cheng et al. [6] and Achibet et al. [1] optimized redirection for distinct primitives and multi-finger contact, while Matthews et al. [24] utilized asymmetric bimanual interaction for interface remapping. Sparse Haptic Proxy uses visibility, proximity, and surface normals but does not maintain reusable load-bearing stacks.

Recent studies have expanded these concepts to various interaction scenarios. Shi et al. [30] introduced ReachPad to map multiple virtual screens to a single physical pad. VoxelHap builds modular proxies [8]. Stacked Retargeting combines redirected walking and hand redirection to broaden fixed-proxy coverage [7]; our method instead reallocates multiple movable proxies while preserving load-bearing support, exposed faces, and future availability in an evolving stack. To maintain the hardware simplicity of passive haptics, researchers have explored optimization-based approaches, including functional optimization for continuous touch [37] and length mismatch resolution [5]. Zhao and Follmer map contact on a fixed 3D boundary. Shape-Aware Haptic Retargeting replaces point distance with hand–target geometric distance for accurate contact [23]; our score instead ranks multiple proxies and sequential placements by stack-wide support, exposure, and future availability. While these works successfully establish retargeting paradigms, extending these optimizations to complex, volumetric spaces remains challenging due to the need for multi-directional physical support. In this paper, we build upon these foundations by proposing a direction-aware retargeting method that utilizes a 3D magnetic proxy system to provide passive haptic feedback for volumetric construction tasks.

Accordingly, cube-based volumetric stacking is used here as a controlled test case for sequential 3D assembly rather than as an end

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

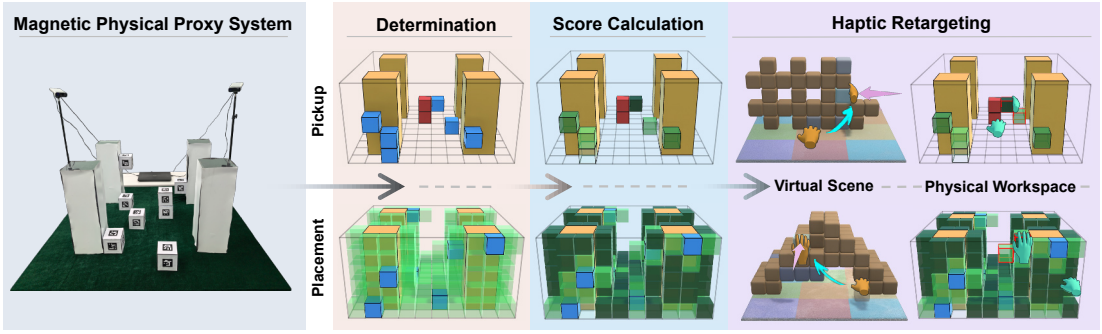


Figure 2: Pipeline of our direction-aware 3D haptic retargeting method.

in itself. Because each placement changes both structural support and the proxies available for later interactions, ignoring this coupling can make a later step infeasible or force an otherwise avoidable reset.

2.2 Perception and Interaction Quality

Haptic retargeting inherently relies on the limits of human perception. Redirection gains must remain within specific detection thresholds to maintain the illusion of presence. Zenner et al. [34] conducted extensive studies to quantify these thresholds for desktop-scale redirection. Zhou et al. [38] expanded this to handheld tools, while Azim et al. [2] investigated the detection of hand redirection from behavioral patterns in mid-air gestures. Beyond interaction with external objects, Bovet et al. [4] examined the impact of visuo-tactile mismatches on embodiment, quantifying the limits of acceptance for distorted mappings during self-contact. In addition to these boundaries, researchers have explored how factors like object geometry [36] and gaze behavior [20] modulate the noticeability of redirection. Feick et al. predict personalized detection limits from movement, gaze, and EEG [9]; we use shared thresholds.

Beyond visual-haptic perceptibility, research has increasingly focused on physical interaction comfort. Feick et al. [10] warned of potential harms such as proprioceptive drift from prolonged redirection. To improve ergonomics, Sun et al. [31] integrated the Rapid Upper Limb Assessment [25] score into the retargeting loop. RULA concerns ergonomic comfort; our score ranks direction-specific support.

Users discriminate 3D fingertip-force directions [27], and force can outweigh geometry during active touch [28]. Bimanual detectability also depends on matching versus opposing offsets [13]; our system is unimanual.

While these studies characterize perceptual limits, ergonomic comfort, and sensitivity to force direction, ensuring that the physical proxy actually provides geometrically accessible and directionally correct rigid support in a cluttered 3D space is less explored. Our method complements this body of work by incorporating Local Exposure and Global Haptic Direction Quality, explicitly guiding users to proxy faces that are geometrically accessible and physically supported opposite to the intended interaction direction.

2.3 Reset Reduction Strategies

Optimized redirection gains may accumulate beyond acceptable thresholds, requiring a system reset to realign the virtual and physical domains. Early research focused on reactive techniques to hide these interruptions, such as utilizing change blindness during blinking [35]. To minimize the immediate recovery effort, Sun et al. [31] further optimized the spatial location of these reset points. Their planar objective combines comfort, travel, gain, and handedness.

To proactively reduce the frequency of these disruptions, researchers have developed various spatial management strategies. Matthews et al. [22] utilized head gaze for explicit target selection to dynamically map physical proxies. Yang et al. [32] optimized placement destinations by minimizing the distance to other virtual objects currently in the scene. More recently, Liu et al. [21] introduced a proxy importance scoring method based on spatial distances and reset probabilities to preserve valuable props. Liu et al. combine proximity, next-step reset risk, and gain in 2D; REACH+ addresses workspace/speed feasibility for an actuated planar device [12].

These spatial management strategies have proven highly effective for planar layouts or sparse proxy configurations. However, managing resets in dense 3D stacking tasks requires further consideration of structural occlusions and proxy availability, as geometric stacking creates complex physical dependencies. To mitigate resets in such volumetric environments, our approach introduces Non-Occupied Proxy Availability. By explicitly evaluating the 3D magnetic attachment states, our method quantitatively evaluates how each interaction alters the availability of physical resources, favoring configurations that preserve proxies for subsequent operations and reduce future reset risk.

3 Method

We propose a direction-aware 3D haptic retargeting method that supports efficient retargeted interaction and directionally supported physical contact in the evaluated cubic-stacking tasks. The difference between our method and existing methods lies in the use of multiple movable physical proxies within a 3D volume to provide passive haptics for virtual objects. This enables efficient interaction in these tasks while providing physical contact supported in the intended interaction direction.

Figure 2 illustrates the pipeline of our method. We first design a magnetic physical proxy system consisting of magnetic bases and multiple proxy cubes, which can be magnetically arranged at various positions within the 3D volume (Sec. 3.1). In each retargeting step, we determine the sets of non-occupied proxies and available placement positions based on their magnetic attachment relationships (Sec. 3.2). Next, we calculate the Haptic-Direction Score for these non-occupied proxies and available placement positions, derived from factors such as spatial adjacency, the quantity of future available proxies, and the statistics of future haptic directions (Sec. 3.3). Finally, the optimal physical proxy or placement position for retargeting is selected based on these scores and the real-time positions of the virtual/real hand, and virtual targets (Sec. 3.4).

3.1 Magnetic Physical Proxy System

We designed a magnetic physical proxy system to facilitate the placement of proxies in 3D space. Figure 3 shows the photos of the systems, highlighting magnetic bases, proxy cubes, the RGB-D cameras, and interaction area. The base is constructed from one or more rectangular acrylic prisms covered with 0.7-mm ferromagnetic receptive sheets, with a bounding box dimension of approximately $80 \times 80 \times 50$ cm. We designed two base configura-

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

tions: High-Plane Base (left) and 4-Pillar Base (right). **High-Plane Base:** A centralized large cuboid (ranging from $40 \times 40 \times 30$ cm to $40 \times 40 \times 50$ cm). This design aims to maximize the continuous horizontal surface area. **4-Pillar Base:** Dispersed vertical pillars (4 pillars, each with a footprint of 20×20 cm and heights ranging from 30 to 50 cm). This design aims to maximize side accessibility and reduce occlusion. The proxies are EVA cubes ($10 \times 10 \times 10$ cm) covered with 2-mm, 600-Gs flexible magnetic sheets on all six faces, which allows them to attach to the base or other cubes. The magnetic cube faces and receptive base require no N/S pole layout.

To track physical proxy cubes during haptic retargeting, we affixed ArUco markers [11] (5×5 cm) to each cube's faces. We deployed two RGB-D cameras to cover an interaction area of $100 \times 100 \times 50$ cm from different viewpoints, with the magnetic base positioned at the center of this interaction area. This setup enables real-time tracking of each cube's position and orientation.

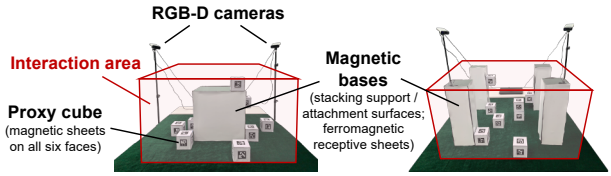


Figure 3: The magnetic physical proxy system: High-Plane Base (left) or 4-Pillar Base (right), 2 RGB-D cameras, 8 proxy cubes.

Before the haptic retargeting session, we perform a system calibration. An ArUco-tagged cube is placed at the bottom-left corner of the physical interaction area. Using ArUco marker tracking, we calibrate the camera positions to align the virtual interaction area with the physical one. Subsequently, the attachment positions of all physical proxies are randomly initialized; in the user study, the same initialization is reused across conditions for each corresponding trial (Sec. 5). Additionally, the user's hand position is acquired in real time using the PICO HMDs. During brief ArUco-marker occlusions in placement, the PICO-tracked hand pose serves as auxiliary information for smoothing the camera-based proxy observations.

3.2 Non-occupied Proxies and Available Placement Positions Determination

We categorize the proxy cubes within the magnetic system into two types: *occupied proxies* and *non-occupied proxies*. A proxy is considered occupied if its removal would cause the other proxies directly attached to it to collapse; otherwise, it is defined as non-occupied. As illustrated in the left image of Figure 4, non-occupied proxies are visualized in blue, while occupied proxies are shown in red. For placement, we evaluate empty voxels within the physical interaction volume. A voxel is designated as an *available placement position* if placing a proxy there results in a successful magnetic attachment to the base or other existing physical proxies. These available placement positions are visualized as transparent green voxels in the right image of Figure 4.

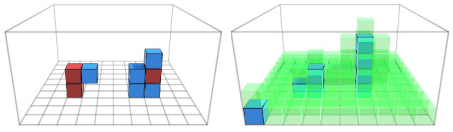


Figure 4: Different types of proxies and placement positions.

3.3 Haptic-Direction Score Calculation

Haptic-Direction Score is a metric designed to quantify the suitability of a non-occupied proxy or an available placement position as the target for the current pickup or placement. A higher score indicates superior operability for the specific proxy or position, i.e., it implies fewer magnetic attachment surfaces, facilitating easier manipulation (pickup or placement), and a structural configuration that enhances the operability of other proxies in subsequent interactions. In contrast, a lower score indicates poor operability or a configuration that hinders the operability of other proxies for future tasks.

By calculating this Haptic-Direction Score, we quantitatively evaluate each non-occupied proxy and available placement position based on Local Exposure, Global Haptic Direction Quality, and Non-occupied Proxy Availability.

For each non-occupied proxy or available placement position P, we calculate the Haptic-Direction Score $H(P)$ by using the product of three scores:

$$H(P) = E(P) \times Q(P) \times A(P) \tag{1}$$

where $E(P)$ represents Local Exposure, $Q(P)$ represents Global Haptic Direction Quality, and $A(P)$ represents Non-occupied Proxy Availability.

Aggregation Rationale. We multiply rather than add because the product gates joint suitability: a weak factor suppresses the score and $Q(P) = 0$ vetoes a candidate, so proximity cannot offset unsupported contact or poor future availability. This matches robust selection. Max-normalizing components preserves ranking. Against an equal-weight sum on the same 100 pairs, the product reduced Reset Rate by 2.68 p.p. and raised Best-/Normal-Touch by $3.81/1.03$ p.p.; Sec. 5.4 ablates each term.

Local Exposure denoted as $E(P)$, quantifies the operability of a non-occupied proxy or available placement position P. A higher score implies that the proxy has fewer magnetic attachment surfaces (i.e., more empty adjacent faces), thereby facilitating easier manipulation and allowing hand approach from multiple angles. This score is calculated as shown in Equation 2:

$$E(P) = \exp(\mathrm{NE}(P) \cdot \lambda) \tag{2}$$

where $\mathrm{NE}(P)$ represents the number of empty faces adjacent to the six cardinal directions of P (with a range of $[1,6]$), and λ is a predefined scaling constant, set to 0.6 in our implementation.

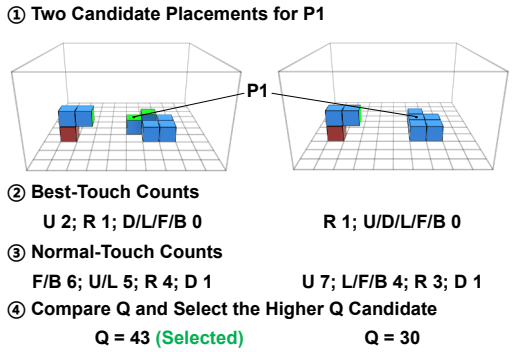


Figure 5: Callouts 1–4 show two P_1 placements, directional best-/normal-touch counts, $Q(P)$, and higher-score selection. U/D/L/R/F/B denote up/down/left/right/front/back. Blue indicates non-occupied proxies, red indicates occupied proxies, and green faces denote best-touch surfaces.

Global Haptic Direction Quality denoted as $Q(P)$ quantifies the collective operability of non-occupied proxies available for subsequent interactions. For a pickup task, this metric evaluates the configuration of all other non-occupied proxies excluding P, whereas for a placement task, it evaluates the configuration including the newly placed P. We define the face opposite to the attachment surface as the *best-touch* surface if the proxy has exactly five empty adjacent faces, while the remaining empty faces are defined as *normal-touch* surfaces. If a proxy has fewer than five empty faces,

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

all empty faces are considered normal-touch surfaces. This score is calculated as shown in Equation 3:

$$Q(P) = \sum_{d \in \mathscr{D}} (BS_d + NS_d) \tag{3}$$

where $\mathscr{D}$ represents the set of six cardinal directions. BS_d and NS_d are the weighted scores for best-touch and normal-touch surfaces oriented toward direction d, defined as:

$$BS_d = \begin{cases} \alpha + \beta \cdot \min(n_{b,d}, 3) & \text{if } n_{b,d} \geq 1 \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

$$NS_d = \min(n_{n,d}, 4) \tag{5}$$

where $n_{b,d}$ and $n_{n,d}$ represent the number of best-touch and normal-touch surfaces oriented toward direction d. α and β are predefined scaling constants, set to 8 and 2 in our implementation.

Figure 5 compares two P_1 placements using directional best-/normal-touch counts. The left candidate contributes $16/14/1$ for up/right/down and 4 for each other direction, totaling $Q(P) = 43$; the right contributes $4/13/1$ and three 4s, totaling 30. Thus, the left is selected because it retains more high-quality approach and support directions for later interactions, not only the immediately contacted face.

Non-occupied Proxy Availability denoted as $A(P)$ quantifies the quantity of non-occupied proxies available for subsequent interactions. In a stacking scenario, interacting with a proxy P (whether pickup or placement) alters the magnetic attachment relationships, which may increase or decrease the number of non-occupied proxies. A higher score implies a larger number of non-occupied proxies remaining in the workspace, thereby enhancing the operability for future tasks. This score is calculated as shown in Equation 6:

$$A(P) = 2^{N_{total}} \tag{6}$$

where N_{total} represents the total number of non-occupied proxies in the 3D volume after the interaction with P.

Parameter Sensitivity. On the same 100 pairs, one-at-a-time tests of $\alpha \in \{4, 8, 12\}$, $\beta \in \{1, 2, 3\}$, and $\lambda \in \{0.3, 0.6, 0.9\}$ produced maximum Reset-/Best-/Normal-Touch deviations of $4.13/6.09/1.97$ p.p. No setting improved all three, so the central defaults $(8, 2, 0.6)$ were retained.

3.4 Haptic Retargeting based on Haptic-Direction Score

During pickup or placement operations, we perform haptic retargeting to determine the subsequent virtual hand position. This process utilizes the Haptic-Direction Scores calculated for each non-occupied proxy and available placement position, combined with the current real hand position, the current virtual hand position, and the information regarding the virtual target object or placement site. The specific process is divided into two steps:

Step 1: Candidate Filtering. We identify feasible targets non-occupied proxies for a pickup task or available positions for a placement task by filtering out candidates that fail to support the intended interaction direction or exceed predefined redirection gain thresholds.

For any candidate, we calculate three 3D redirection gains: horizontal angular warp, vertical angular warp, and translational gain [34]. A candidate is valid only if it satisfies two conditions:

(1) *Directional Alignment*: The physical candidate must share the same operable approach direction as the virtual target without obstruction.

(2) *Gain Thresholds*: The calculated gains must fall within acceptable limits. While classical 3D retargeting enforces strict imperceptible bounds ($\pm 4.5^\circ$) [34], recent planar retargeting studies demonstrate that relaxing the angular threshold to $\pm 15^\circ$ and the translational gain to $0.83 \sim 1.2$ significantly reduces system resets while preserving interaction comfort [21, 31]. To balance interaction continuity and physical reachability in our volumetric tasks, we systematically extend these relaxed parameters to 3D space: setting both the horizontal and vertical angular thresholds to $\pm 15^\circ$, and the translational gain threshold to $[0.83, 1.2]$. This unified 3D relaxation is used for our evaluated continuous stacking tasks, which heavily involve "reach-to-place" actions, a kinematic phase where users inherently exhibit a higher tolerance for visuo-haptic mismatches [33].

Step 2: Reset Mechanism and Optimal Target Selection.

If the filtered candidate set is empty, we guide both the virtual and real hands to a calculated reset point to re-evaluate target availability based on the updated positions. To determine the optimal reset point, we perform a volumetric search expanding outward layer-by-layer from the current real hand position, treating voxel centers as potential reset points. A voxel is added to the reset candidate set only if it satisfies two conditions:

(1) *Transition Compliance*: The redirection gains required to move the hand from the current position to this reset point fall within permissible thresholds.

(2) *Subsequent Feasibility*: Simulating the hand at this reset point allows the candidate filtering process (Step 1) to identify a non-empty set of valid targets.

The search terminates once N valid reset candidates are collected ($N = 5$ in our implementation). We then select the candidate with the highest score S_r, calculated as:

$$S_r = k \cdot H^* - d_{h,r} - d_{r,t} \tag{7}$$

where H^* is the maximum Haptic-Direction Score achievable from the candidate reset point, $d_{h,r}$ is the Euclidean distance from the current real hand position to the reset point (in meters), $d_{r,t}$ is the distance from the reset point to the corresponding physical target (in meters), and k is a scaling constant (set to 100) used to balance the scoring metric with the spatial displacement.

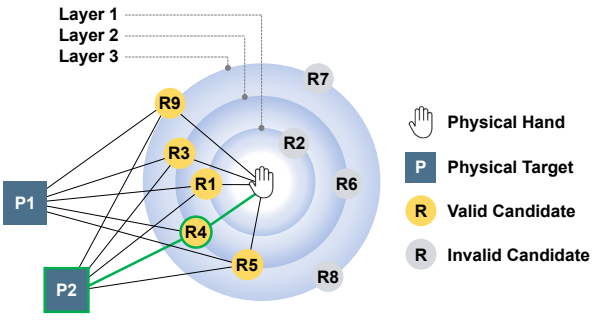


Figure 6: Determination of the optimal reset point.

Figure 6 presents a schematic illustrating the determination of the optimal reset point and the selection of the best physical target. Our method systematically explores feasible reset points from near to far. While ensuring that the physical movement distance required for the reset remains minimal, it continuously seeks targets that yield a higher Haptic-Direction Score, thereby effectively optimizing the spatial layout of the physical proxies. Centered on the real hand position, the search for valid reset candidates begins in the nearest region (Layer 1). For example, assuming R1 and R2 are candidates on Layer 1, R1 satisfies both the transition compliance and subsequent feasibility (valid candidate), whereas R2 fails to meet these constraints (invalid candidate). The search progressively expands to layers with larger radii until five valid candidates (e.g., R1, R3, R4, R5, and R9) are identified. For each valid candidate, S_r is calculated based on the distance from the real hand to the reset point ($d_{h,r}$), the distance from the reset point to a potential target P ($d_{r,t}$), and the Haptic-Direction Score of that corresponding

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

target P. The candidate yielding the highest score (R4) is selected as the optimal reset point. Ultimately, the physical target offering the highest Haptic-Direction Score reachable from R4 (P2) is determined as the final destination for both the virtual and real hands.

From the filtered candidate set (or the newly evaluated valid targets after a reset), we select the candidate proxy or candidate placement position with the highest Haptic-Direction Score as the optimal physical target in the pickup and placement process.

4 Simulation Experiment: Selecting Base of Magnetic Physical Proxy System

A simulation experiment was designed to determine the base shape and parameters that minimize reset rates while maximizing the Best-Touch and Normal-Touch Ratios.

4.1 Experiment Design

The simulation task involved transforming a stack of virtual blocks from a random source configuration (*Source*) into a distinct target configuration (*Target*), as illustrated in Figure 7(a). We generated 100 unique Source–Target pairs in a $10 \times 10 \times 5$ voxel grid from 200 distinct, connected, supported configurations. Each used a random sequence preserving connectivity and support at every step; all base conditions used identical pairs and sequences.

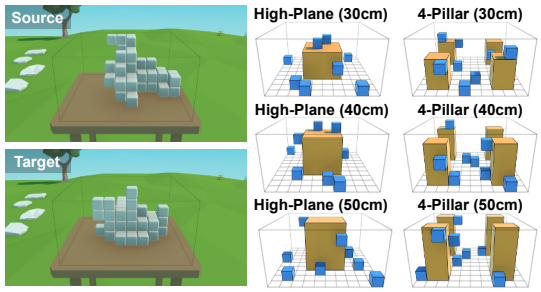


Figure 7: Scenes of the simulation experiment: (a) examples of the virtual source and target scenes; (b) simulated physical magnetic bases of different shapes and sizes.

We conducted a comparative analysis across three base conditions: (1) *No Base*, (2) *High-Plane Base*, and (3) *4-Pillar Base*. The operational volume for both workspaces was $100 \times 100 \times 50$ cm. To determine the optimal vertical support, we tested both High-Plane and 4-Pillar configurations at three heights: 30, 40, and 50 cm. These six physical base configurations are visualized in Figure 7(b).

We evaluate the performance using three quantitative metrics: Reset Rate, Best-Touch Ratio, and Normal-Touch Ratio.

- *Reset Rate* (↓): Indicates the continuity of task execution. It is defined as the percentage of interaction steps (where the pickup and placement of each cube are treated as two independent steps) that necessitate a forced reset operation. A lower rate implies fewer task interruptions and a more seamless user experience.
- *Best-Touch Ratio* (↑): Indicates the consistency of optimal haptic feedback. It is defined as the percentage of interaction steps where the selected proxy (or placement position) offers a "best-touch" surface configuration. This specifically refers to cases where the face opposite to the interaction direction is physically supported by the base or another block, while the remaining five faces are empty. A higher ratio indicates that more interactions have physical backing opposite the intended direction and five exposed faces.
- *Normal-Touch Ratio* (↑): Indicates the baseline consistency of geometric alignment. It is defined as the percentage of interaction steps where the selected proxy offers a "normal-touch" configuration. While these configurations may lack the ideal support or full isolation of a best-touch scenario, they fundamentally guarantee that the physical interaction direction is valid. Crucially, a normal-touch state ensures that the user's hand trajectory aligns with the physical environment, preventing geometric conflicts such as the hand penetrating other proxies (clipping) during the virtual interaction.

We analyzed 100 matched observations using one-way repeated-measures ANOVA, Greenhouse–Geisser correction when Mauchly's test rejected sphericity [14], and 21-pair Bonferroni-adjusted paired tests ($\alpha = 0.05$); analyses used unrounded data and report percentage-point changes.

4.2 Results & Discussion

Table 1 presents the Reset Rate comparisons. Overall, adding a physical base reduced resets relative to No Base (72.84%): the High-Plane configurations averaged 67.23%, whereas the 4-Pillar configurations averaged 64.74%. The 50-cm 4-Pillar Base produced the lowest Reset Rate (63.92%), 8.92 percentage points below No Base ($p < 0.001$, $d = 2.34$). The progressive decrease across height and base type is consistent with the number and spatial distribution of valid placement positions. Four separated anchors reduce the risk that all useful support is concentrated in one region, while the 50-cm extent makes more of the vertical workspace available than the 30- and 40-cm variants. Together, these properties leave more feasible configurations for subsequent actions and help explain why the longest 4-Pillar design required the fewest simulated resets.

Table 1: Reset Rate (%). $F(4.06, 401.65) = 63.95$, $p < 0.001$.

Condition	Avg ± std. dev.	Reduction vs. No Base (p.p.)	p	Cohen's d	Effect size
No Base (CC)	72.84 ± 4.12	-	-	-	-
High-Plane (30 cm) (EC1)	68.21 ± 3.86	4.63	$< 0.001^*$	1.16	Large
High-Plane (40 cm) (EC2)	67.11 ± 3.73	5.72	$< 0.001^*$	1.45	Very Large
High-Plane (50 cm) (EC3)	66.36 ± 3.92	6.48	$< 0.001^*$	1.61	Very Large
4-Pillar (30 cm) (EC4)	65.72 ± 3.65	7.12	$< 0.001^*$	1.83	Very Large
4-Pillar (40 cm) (EC5)	64.59 ± 3.53	8.25	$< 0.001^*$	2.15	Huge
4-Pillar (50 cm) (EC6)	63.92 ± 3.48	8.92	$< 0.001^*$	2.34	Huge

Table 2: Best-Touch Ratio (%). $F(3.94, 390.28) = 71.56$, $p < 0.001$.

Condition	Avg ± std. dev.	Increase vs. No Base (p.p.)	p	Cohen's d	Effect size
No Base	53.64 ± 5.82	-	-	-	-
High-Plane (30 cm)	58.71 ± 5.94	5.07	$< 0.001^*$	0.86	Large
High-Plane (40 cm)	59.83 ± 6.12	6.19	$< 0.001^*$	1.03	Large
High-Plane (50 cm)	60.92 ± 5.88	7.28	$< 0.001^*$	1.24	Very Large
4-Pillar (30 cm)	62.87 ± 6.31	9.23	$< 0.001^*$	1.52	Very Large
4-Pillar (40 cm)	64.39 ± 6.47	10.75	$< 0.001^*$	1.74	Very Large
4-Pillar (50 cm)	65.81 ± 6.73	12.17	$< 0.001^*$	1.93	Very Large

The largest base-related change occurred in the Best-Touch Ratio (Table 2). It increased from 53.64% with No Base to averages of 59.82% for High-Plane and 64.36% for 4-Pillar configurations. The 50-cm 4-Pillar Base reached 65.81%, 12.17 percentage points above No Base ($p < 0.001$, $d = 1.93$). This pattern is consistent with the geometric exposure of the supports. A narrow pillar can provide rigid backing without occluding as many lateral faces as a solid plane, and additional height allows proxies to be distributed across vertical levels rather than densely packed. More candidates can therefore retain the five-empty-face configuration used by the Best-Touch criterion, making an optimally supported direction available more often.

Table 3 reports the Normal-Touch Ratio. The No Base condition averaged 92.17%, compared with 92.82% across High-Plane configurations and 94.14% across 4-Pillar configurations. The 50-cm 4-Pillar Base again had the highest mean (94.46%), 2.29 percentage points above No Base ($p < 0.001$, $d = 1.23$). Only the three 4-Pillar–No-Base contrasts remained significant after full-family adjustment, so the small High-Plane increases should not be interpreted as reliable improvements. The stronger 4-Pillar pattern is consistent with dispersed contact surfaces at several horizontal locations and heights: a requested cardinal direction is less likely to

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

be blocked everywhere at once. This broader distribution preserves feasible normal-touch opportunities even when a best-touch surface is unavailable.

Table 3: Normal-Touch Ratio (%). $F(2.76, 273.00) = 27.03$, $p < 0.001$.

Condition	Avg ± std. dev.	Increase vs. No Base (p.p.)	p	Cohen's d	Effect size
No Base	92.17 ± 2.08	-	-	-	-
High-Plane (30 cm)	92.58 ± 1.94	0.41	1.000	0.20	Small
High-Plane (40 cm)	92.83 ± 1.88	0.66	0.199	0.33	Small
High-Plane (50 cm)	93.05 ± 1.76	0.88	0.077	0.45	Small
4-Pillar (30 cm)	93.85 ± 1.72	1.68	$< 0.001^*$	0.86	Large
4-Pillar (40 cm)	94.12 ± 1.63	1.95	$< 0.001^*$	1.03	Large
4-Pillar (50 cm)	94.46 ± 1.51	2.29	$< 0.001^*$	1.23	Very Large

Simulation results confirm that the 4-Pillar Base (50 cm) simultaneously has the highest measured Normal-Touch and Best-Touch Ratios and the lowest Reset Rate among the tested bases. Compared to a solid High-Plane Base, the distributed pillar layout provides more side access and more candidate support directions within the evaluated layouts. Consequently, 4-Pillar Base (50 cm) was adopted in the subsequent user study.

5 User Study

To evaluate the performance of our direction-aware 3D haptic retargeting method, we conducted a formal within-subjects user study.

5.1 Participants and Apparatus

We recruited 20 participants (12 males and 8 females, aged 22-28, mean=24.5) from the university. All participants had normal or corrected-to-normal vision and no history of motor or balance disorders. 15 of them had prior experience with VR interfaces. The research was performed under the oversight of the University's Ethics Committee. Consent from the human subjects in the research was obtained. The hardware setup consisted of a PICO VR headset connected to a PC via streaming, utilizing SteamVR and OpenXR for hand tracking. The physical workspace was monitored by two calibrated RGB-D cameras to track the 6-DOF poses of the physical proxies covered with ArUco markers [11] in real-time. The magnetic physical proxy system utilized the optimized configuration determined in the simulation experiment: 4-Pillar Base (50 cm).

5.2 Experimental Design

Task This task required participants to construct transformations from source to target shapes within a $5 \times 7 \times 3$ m virtual game room scenario, on a $100 \times 100 \times 50$ cm open area. The experimental tasks for the three sessions are illustrated in Figure 8: Session 1 involved rebuilding stairs (30 blocks) into a fence; Session 2 rebuilt a bench (60 blocks) into a bridge; and Session 3 rebuilt a tetris pile (100 blocks, randomized for each trial) into a boxing ring.

To ensure fair matched comparisons, each trial used a randomly generated fixed sequence of valid pickup and placement operations, and the same sequence was reused across all seven conditions for the corresponding trial. In Session 3, the randomized Tetris source layout was likewise matched across conditions for the corresponding trial. During tasks, the system highlighted target blocks in orange for selection and displayed a semi-transparent "ghost block" at the placement position to guide user actions. The source positions of the physical proxies were randomized for each trial and matched across conditions.

Conditions This study includes three planar control conditions adapted to 3D (CC1, CC2, CC3) and an experimental condition (EC) employing our method, plus three ablations removing $E(P)$, $Q(P)$, or $A(P)$ (w/o E/Q/A). All participants completed these seven conditions in a balanced order. CC1 is our 3D adaptation of the optimal mapping strategy [22] as described by Liu et al. [21]. It dynamically selects the physical proxy based on the closest distance to the virtual target and determines the placement position by preserving their relative positional relationship. CC2 is our 3D adaptation of the Retargeting Placement approach [32], primarily optimizing placement positions. When a user places a virtual object, the system selects the physical placement position minimizing the total sum of distances to other potential future virtual target positions. CC3 is our 3D adaptation of the Proxy Importance-based Haptic Retargeting method [21]. It calculates an importance score for each physical proxy by comprehensively considering factors such as proxy proximity and avoidance probability of repositioning. The system selects the proxy with the lowest importance during pickup and the placement position with the highest importance during placement. Since CC1-CC3 are originally 2D methods based on planar proxy distribution, we extended them to adapt to the 3D space: the set of candidate proxies was expanded to include non-occupied proxies in 3D space; the set of candidate placement positions was expanded to include available placement positions in 3D space; and the calculation of retargeting gains was generalized from 2D to 3D. No passive-haptic comparator targets 3D stacking, so we use the closest adaptable methods. Their objectives omit vertical support, evolving occupancy, and exposed faces; results characterize this task, not their general performance.

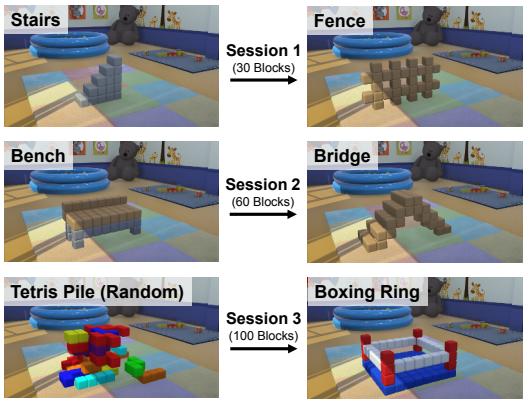


Figure 8: The three sessions of the block-stacking task in the user study, showing the transformation from source to target with increasing block counts.

Procedure Participants were first briefed on the concept of haptic retargeting and the task requirements. They performed a 5-minute training session to familiarize themselves with the PICO headset and the block-stacking mechanics. The main experiment followed a counterbalanced order for the seven conditions. Between conditions and sessions, participants took a 3-minute break to minimize fatigue. After completing each condition, participants filled out a subjective questionnaire.

5.3 Metrics

We evaluated participant performance using a comprehensive set of objective and subjective metrics. These metrics expand upon those used in the simulation (Reset Rate, Best-Touch Ratio, and Normal-Touch Ratio) by adding Reset, Efficiency, and User Experience dimensions suitable for human-in-the-loop evaluation, as detailed below.

Reset:
- *Reset Time* (↓): The total time consumed by reset operations.
- *Reset Distance* (↓): The total physical distance traveled by the user's hand during reset operations.

Efficiency:
- *Task Completion Time* (↓): The total duration required to complete the construction task, excluding training time.
- *Average Hand Movement Distance* (↓): The average physical distance traveled by the user's hand per operation.

User Experience:
- *NASA-TLX Score* (↓): The perceived workload assessed during the task [15, 16].

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

- *Usability Score* (↑): The system usability evaluated via a 9-point questionnaire [18] covering six key dimensions: intuitiveness, efficiency, accuracy, naturalness, satisfaction, and easiness.

5.4 Results

We used one-way RM-ANOVA with Greenhouse–Geisser correction when sphericity failed [14], followed by two-sided paired t-tests with 21-pair Bonferroni correction per outcome ($\alpha = 0.05$). Tests used unrounded data; tables give native-unit changes, F/df, and repeated-measures Cohen's d [29]; all pairs appear in Supplementary Tables S1–S23.

5.4.1 Reset Analysis

Against the three planar controls adapted to 3D (CC1–CC3), EC had the lowest Reset Rate (61.37%; $-13.65/11.06/6.77$ p.p.), Reset Distance (12.85 m; $-32.45/29.80/18.60$ m), and Reset Time (85.25 s; $-101.25/86.90/43.15$ s; all $p < 0.001$; Tables 4–6). Thus, both reset frequency and recovery cost declined. Removing $E(P)/A(P)$ increased Rate by $7.72/10.94$ p.p., Distance by $7.48/9.68$ m, and Time by $40.70/54.79$ s (all adjusted $p \leq 0.0053$); removing $Q(P)$ produced no significant reset-metric change.

Table 4: Reset Rate (%). $F(6,114) = 21.58$, $p < 0.001$.

Condition	Avg ± std. dev.	Change from EC (p.p.)	p	Cohen's d	Effect size
EC	61.37 ± 3.28	-	-	-	-
w/o E	69.09 ± 5.40	7.72	$< 0.001^*$	1.70	Very Large
w/o Q	62.57 ± 5.67	1.20	1.000	0.26	Small
w/o A	72.31 ± 6.10	10.94	$< 0.001^*$	2.20	Huge
CC1	75.02 ± 4.91	13.65	$< 0.001^*$	3.32	Huge
CC2	72.43 ± 4.65	11.06	$< 0.001^*$	2.75	Huge
CC3	68.14 ± 4.12	6.77	$< 0.001^*$	1.82	Very Large

Table 5: Reset Distance (m). $F(3.61, 68.56) = 119.51$, $p < 0.001$.

Condition	Avg ± std. dev.	Change from EC (m)	p	Cohen's d	Effect size
EC	12.85 ± 2.45	-	-	-	-
w/o E	20.33 ± 5.14	7.48	$< 0.001^*$	1.89	Very Large
w/o Q	13.88 ± 3.57	1.03	1.000	0.33	Small
w/o A	22.53 ± 8.02	9.68	0.002^*	1.69	Very Large
CC1	45.30 ± 6.25	32.45	$< 0.001^*$	6.85	Huge
CC2	42.65 ± 5.80	29.80	$< 0.001^*$	6.62	Huge
CC3	31.45 ± 4.15	18.60	$< 0.001^*$	5.43	Huge

Table 6: Reset Time (s). $F(3.67, 69.80) = 35.56$, $p < 0.001$.

Condition	Avg ± std. dev.	Change from EC (s)	p	Cohen's d	Effect size
EC	85.25 ± 15.35	-	-	-	-
w/o E	125.95 ± 37.97	40.70	0.005^*	1.40	Very Large
w/o Q	88.88 ± 23.63	3.63	1.000	0.18	Very Small
w/o A	140.04 ± 45.45	54.79	0.001^*	1.62	Very Large
CC1	186.50 ± 28.80	101.25	$< 0.001^*$	4.35	Huge
CC2	172.15 ± 26.45	86.90	$< 0.001^*$	3.98	Huge
CC3	128.40 ± 22.60	43.15	$< 0.001^*$	2.24	Huge

5.4.2 Efficiency Analysis

Table 7: Task Completion Time (s). $F(6,114) = 27.06$, $p < 0.001$.

Condition	Avg ± std. dev.	Change from EC (s)	p	Cohen's d	Effect size
EC	385.40 ± 32.50	-	-	-	-
w/o E	438.02 ± 38.35	52.62	0.002^*	1.48	Very Large
w/o Q	391.29 ± 38.38	5.89	1.000	0.17	Very Small
w/o A	463.32 ± 29.10	77.92	$< 0.001^*$	2.53	Huge
CC1	498.85 ± 48.60	113.45	$< 0.001^*$	2.73	Huge
CC2	485.20 ± 45.40	99.80	$< 0.001^*$	2.52	Huge
CC3	422.65 ± 38.30	37.25	0.152	1.05	Large

Table 8: Average Hand Movement Distance (m). $F(3.45, 65.52) = 16.91$, $p < 0.001$.

Condition	Avg ± std. dev.	Change from EC (m)	p	Cohen's d	Effect size
EC	0.94 ± 0.12	-	-	-	-
w/o E	1.01 ± 0.16	0.07	1.000	0.53	Medium
w/o Q	0.97 ± 0.21	0.03	1.000	0.19	Very Small
w/o A	1.09 ± 0.21	0.15	0.275	0.85	Large
CC1	1.28 ± 0.18	0.34	$< 0.001^*$	2.21	Huge
CC2	1.34 ± 0.19	0.40	$< 0.001^*$	2.48	Huge
CC3	1.09 ± 0.13	0.15	$< 0.001^*$	1.15	Large

Against the 3D-adapted CC1/CC2/CC3 controls, EC task time was 385.40 s, lower by $113.45/99.80/37.25$ s ($p < 0.001/ < 0.001/0.152$; Table 7); only the CC3 difference was nonsignificant. Hand Movement Distance was 0.94 m per interaction, $0.34/0.40/0.15$ m lower (all $p < 0.001$; Table 8), indicating less travel, not necessarily less exertion. Removing $E(P)/A(P)$ added $52.62/77.92$ s (adjusted $p = 0.002/ < 0.001$); no ablation distance contrast was significant. Error rate and exertion were not evaluated.

5.4.3 Haptic Quality Analysis

These ratios are system-defined geometry, not perceived realism. Against the three planar controls adapted to 3D (CC1–CC3), EC scored 46.85% Best-Touch ($+24.05/25.70/18.45$ p.p.) and 85.24% Normal-Touch ($+6.59/5.44/3.74$ p.p.; all $p < 0.001$; Tables 9–10). Removing $Q(P)$ reduced them by $15.64/12.07$ p.p. (both adjusted $p < 0.001$); $E(P)/A(P)$ effects were nonsignificant.

Table 9: Best-Touch Ratio (%). $F(3.34, 63.55) = 49.82$, $p < 0.001$.

Condition	Avg ± std. dev.	Change from EC (p.p.)	p	Cohen's d	Effect size
EC	46.85 ± 5.85	-	-	-	-
w/o E	40.81 ± 9.32	−6.04	0.154	0.76	Medium
w/o Q	31.21 ± 10.45	−15.64	$< 0.001^*$	1.80	Very Large
w/o A	45.56 ± 10.54	−1.29	1.000	0.15	Very Small
CC1	22.80 ± 4.10	−24.05	$< 0.001^*$	4.75	Huge
CC2	21.15 ± 3.95	−25.70	$< 0.001^*$	5.14	Huge
CC3	28.40 ± 4.80	−18.45	$< 0.001^*$	3.45	Huge

Table 10: Normal-Touch Ratio (%). $F(3.18, 60.46) = 11.59$, $p < 0.001$.

Condition	Avg ± std. dev.	Change from EC (p.p.)	p	Cohen's d	Effect size
EC	85.24 ± 3.55	-	-	-	-
w/o E	81.10 ± 6.00	−4.14	0.145	0.82	Large
w/o Q	73.17 ± 9.16	−12.07	$< 0.001^*$	1.71	Very Large
w/o A	82.79 ± 5.77	−2.45	1.000	0.50	Medium
CC1	78.65 ± 4.25	−6.59	$< 0.001^*$	1.68	Very Large
CC2	79.80 ± 4.10	−5.44	$< 0.001^*$	1.41	Very Large
CC3	81.50 ± 3.95	−3.74	$< 0.001^*$	0.99	Large

5.4.4 User Experience Analysis

All NASA–TLX outcomes and five usability outcomes showed condition effects ($p < 0.001$); Naturalness did not ($F(3.72, 70.66) = 1.85$, $p = 0.133$). Full F/df are in Supplementary Tables S11–S23.

Against the three planar controls adapted to 3D (CC1–CC3), 37/39 adjusted contrasts favored EC; only Accuracy and Naturalness versus CC3 were nonsignificant (Figure 9). EC overall NASA–TLX was 25.54 versus CC3's 41.25, and frustration was 16.75 versus CC1/CC3's $79.25/42.25$. These results are consistent with fewer interruptions and predictable support, not mechanism isolation. Without $E/Q/A$, overall NASA–TLX was $35.50/32.58/34.63$ (adjusted $p < 0.001/0.009/ < 0.001$ versus EC).

EC exceeded the controls in 16/18 adjusted usability comparisons, reaching 6.10 easiness, 6.05 efficiency, and 6.00 satisfaction (CC3: 4.75). Only four of 18 favorable ablation contrasts were significant, so this evidence is weaker.

6 Discussion

6.1 Interpretation and Comparator Scope

Regarding the frequency of system resets, the performance differences highlight the structural challenges of transitioning from planar retargeting to 3D construction tasks. In our 3D adaptations, CC1 employs a greedy distance metric, selecting the nearest physical proxy. This can scatter proxies and leave insufficient vertical height for later stacking. CC2 optimizes placement by minimizing the total distance to all other virtual objects currently in the scene. In 3D, this can centralize proxies and turn supporting blocks into occupied resources. CC3 introduces proxy importance based on distances and reset probabilities. Its planar coverage objective can limit vertical accumulation and does not model evolving support dependencies. In contrast, our method synergizes the

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

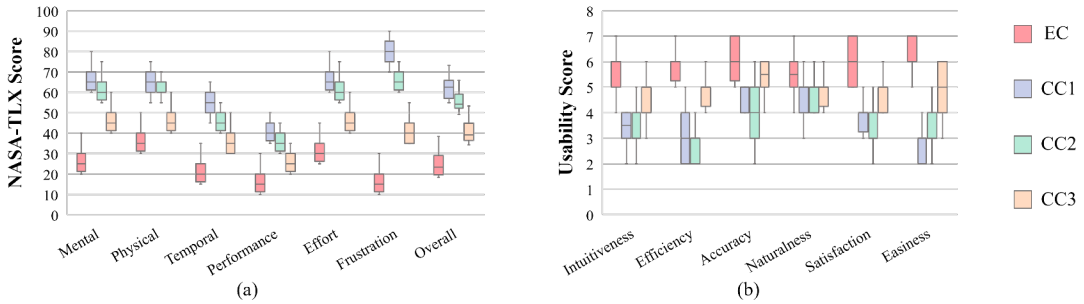


Figure 9: Subjective results for EC and three planar controls adapted to 3D (CC1–CC3): (a) NASA–TLX (lower is better); (b) usability (higher is better).

computation of Non-occupied Proxy Availability with the structural advantages of 4-Pillar Base. Separated anchors reduce clustering, while attachment-aware availability discourages scattering and preserves reusable placements. Because CC1–CC3 are planar methods adapted to 3D for this study, the comparison tests 3D-aware support and occupancy in the evaluated task, not the original methods' general robustness.

In their 3D adaptations, CC1, CC2, and CC3 optimize proximity, placement distance, and proxy importance, respectively. However, the increased frequency of forced resets in 3D stacking tasks can offset these benefits. Our adaptations of these control conditions traditionally rely on fixed reset points. When a reset is unavoidable, EC employs a dynamic reset mechanism. By conducting a volumetric search expanding outward from the current real hand position, it calculates an optimal, localized reset point that minimizes the immediate recovery distance while guaranteeing geometric feasibility for the next step. This dynamic localization translates directly into faster task execution and lower reset-recovery movement under matched sequences. Error rate and exertion were not measured; efficiency is limited to time, movement, reset frequency, and recovery cost.

The three planar controls adapted to 3D (CC1–CC3) utilize distance-based optimizations that efficiently guide the user's hand to a spatial coordinate. However, in a 3D environment, pick-and-place tasks often require multi-directional approaches, such as picking up a proxy from the side or placing a proxy horizontally against an existing structure. Because the control conditions do not explicitly model the orientation of supported and exposed faces, providing a "best-touch" surface for lateral interactions relies primarily on chance; this typically occurs during standard vertical actions where an underlying proxy naturally provides support. Conversely, our method explicitly addresses 3D directionality by calculating Local Exposure and Global Haptic Direction Quality. It actively evaluates the available empty faces and prioritizes candidates that provide rigid physical support precisely opposite to the intended virtual interaction direction, which increased the measured Best-Touch rate. These geometric-support ratios do not measure perceived realism; their decrease without $Q(P)$ supports the directional term.

By resolving the spatial distribution challenges inherent in 3D stacking, streamlining hand movement with dynamic resets, and prioritizing consistent, multi-directional haptic feedback, our proposed method was accompanied by lower NASA–TLX workload ratings and higher usability scores. This demonstrates that the direction-aware scoring framework is highly effective for the evaluated cubic-stacking tasks; broader 3D environments require further validation.

Volumetric stacking couples exposure, load-bearing occupancy, and future availability beyond prior proximity/coverage work [22, 32, 21, 23]; the magnetic base is one support realization, and equivalent connectors remain untested.

6.2 Implications and Limitations

First, the physical base or existing proxies may obstruct the user's hand trajectory. Reactive red voxels warn of impending collisions. Future work aims to proactively visualize optimal, collision-free trajectories using path-planning algorithms. During marker occlusion, PICO hand pose assists camera observations; residual error was unquantified.

Magnetic attraction, detachment resistance, and snap-off were not measured or isolated and may independently influence perception or effort. These connector forces are distinct from the intended geometric and support cues; controlled comparisons across connector types remain future work.

With uniform 1:1 cubic proxies, the Haptic-Direction Score assumes an orthogonal six-face topology. Thus, scope is limited to axis-aligned stacks, six cardinal directions, and at most five exposed faces. This excludes scale changes, arbitrary rotations, and non-convex geometry. We plan to generalize the scoring framework to accommodate variable scales, curved surfaces, and complex polyhedra for broader virtual scenarios. The quasi-static rigid-stack assumption also limits moving or non-rigid/deformable objects; changing support needs dynamic models, online estimation, and revalidation.

Finally, our method is primarily optimized for uni-manual interaction. Bimanual use could reserve the stabilizing proxy, faces, and workspace before rescoring; this remains unimplemented.

7 Conclusion

We have proposed a direction-aware 3D haptic retargeting method in VR. Specifically, we designed a magnetic physical proxy system and introduced the concept of non-occupied proxies. By calculating the associated Haptic-Direction Scores, we quantitatively evaluated candidate proxies and candidate placement positions to select the optimal targets for performing pickup, placement, and reset operations. The results of the user study demonstrate that, compared to three planar methods adapted to 3D, our method can significantly reduce the reset rate while reducing task-completion time and hand movement distance and increasing the Best-Touch and Normal-Touch ratios in the evaluated cubic-stacking tasks.

Acknowledgments

This work is supported by the National Natural Science Foundation of China through Projects U25A20441, 62507002 and 62372026, the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China under Grant JYB2025XDXM610, the New Generation Artificial Intelligence–National Science and Technology Major Project 2025ZD0124000, and the Fundamental Research Funds for the Central Universities.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] M. Achibet, B. Le Gouis, M. Marchal, P.-A. Leziart, F. Argelaguet, A. Girard, A. Lécuyer, and H. Kajimoto. Flexifingers: Multi-finger interaction in vr combining passive haptics and pseudo-haptics. In *2017 IEEE Symposium on 3D User Interfaces (3DUI)*, pp. 103–106. IEEE, 2017. 2

[2] M. A. R. Azim, Z. Su, and S. Heo. Your hands can tell: Detecting redirected hand movements in virtual reality. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pp. 1–14, 2025. 3

[3] M. Azmandian, M. Hancock, H. Benko, E. Ofek, and A. D. Wilson. Haptic retargeting: Dynamic repurposing of passive haptics for enhanced virtual reality experiences. In *Proceedings of the 2016 chi conference on human factors in computing systems*, pp. 1968–1979, 2016. 2

[4] S. Bovet, H. G. Debarba, B. Herbelin, E. Molla, and R. Boulic. The critical role of self-contact for embodiment in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 24(4):1428–1436, 2018. 3

[5] J. Chen, Q. Zou, Y. Chang, S. Oh, and S. Je. Extendlibur: Dynamic haptic retargeting for length-mismatched proxies in co-located vr. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–13, 2025. 2

[6] L.-P. Cheng, E. Ofek, C. Holz, H. Benko, and A. D. Wilson. Sparse haptic proxy: Touch feedback in virtual environments using a general passive prop. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, pp. 3718–3728, 2017. 2

[7] A. Clarence, J. Knibbe, M. Cordeil, and M. Wybrow. Stacked retargeting: Combining redirected walking and hand redirection to expand haptic retargeting's coverage. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2024. 2

[8] M. Feick, C. Biyikli, K. Gani, A. Wittig, A. Tang, and A. Krüger. Voxelhap: A toolkit for constructing proxies providing tactile and kinesthetic haptic feedback in virtual reality. In *Proceedings of the 36th annual ACM symposium on user interface software and technology*, pp. 1–13, 2023. 2

[9] M. Feick, K. P. Regitz, L. Gehrke, A. Zenner, A. Tang, T. P. Jungbluth, M. Rekrut, and A. Krüger. Predicting the limits: Tailoring unnoticeable hand redirection offsets in virtual reality to individuals' perceptual boundaries. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–13, 2024. 3

[10] M. Feick, K. P. Regitz, and A. Zenner. Delusionized? potential harms of proprioceptive manipulations through hand redirection in virtual reality. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–9, 2025. 3

[11] S. Garrido-Jurado, R. Muñoz-Salinas, F. J. Madrid-Cuevas, and M. J. Marín-Jiménez. Automatic generation and detection of highly reliable fiducial markers under occlusion. *Pattern Recognition*, 47(6):2280–2292, 2014. 4, 7

[12] E. J. Gonzalez, P. Abtahi, and S. Follmer. Reach+ extending the reachability of encountered-type haptics devices through dynamic redirection in vr. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*, pp. 236–248, 2020. 3

[13] E. J. Gonzalez and S. Follmer. Investigating the detection of bimanual haptic retargeting in virtual reality. In *Proceedings of the 25th ACM Symposium on Virtual Reality Software and Technology*, pp. 1–5, 2019. 3

[14] S. W. Greenhouse and S. Geisser. On methods in the analysis of profile data. *Psychometrika*, 24(2):95–112, 1959. 6, 8

[15] S. G. Hart. Nasa-task load index (nasa-tlx); 20 years later. In *Proceedings of the human factors and ergonomics society annual meeting*, vol. 50, pp. 904–908. Sage publications Sage CA: Los Angeles, CA, 2006. 7

[16] S. G. Hart and L. E. Staveland. Development of nasa-tlx (task load index): Results of empirical and theoretical research. In *Advances in psychology*, vol. 52, pp. 139–183. Elsevier, 1988. 7

[17] B. E. Insko. *Passive haptics significantly enhances virtual environments*. The University of North Carolina at Chapel Hill, 2001. 1

[18] H. Kim, G. Lee, and M. Billinghurst. A non-linear mapping technique for bare-hand interaction in large virtual environments. In *Proceedings of the Annual Meeting of the Australian Special Interest Group for Computer Human Interaction*, pp. 53–61, 2015. 8

[19] L. Kohli, M. C. Whitton, and F. P. Brooks. Redirected touching: Training and adaptation in warped virtual spaces. In *2013 IEEE Symposium on 3D User Interfaces (3DUI)*, pp. 79–86. IEEE, 2013. 1, 2

[20] Z. Li, Y. Ji, R. Chen, T. Liu, Y. Wang, Y. Shi, and Y. Yan. Modeling the impact of visual stimuli on redirection noticeability with gaze behavior in virtual reality. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pp. 1–18, 2025. 3

[21] Z. Liu, J. Wu, L. Wang, X. Li, and S. K. Im. Proxy importance based haptic retargeting with multiple props in vr. *IEEE Transactions on Visualization and Computer Graphics*, 2024. 2, 3, 5, 7, 9

[22] B. J. Matthews and R. T. Smith. Head gaze target selection for redirected interaction. In *SIGGRAPH Asia 2019 XR*, pp. 13–14. 2019. 2, 3, 7, 9

[23] B. J. Matthews, B. H. Thomas, G. S. Von Itzstein, and R. T. Smith. Shape aware haptic retargeting for accurate hand interactions. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 625–634. IEEE, 2022. 2, 9

[24] B. J. Matthews, B. H. Thomas, S. Von Itzstein, and R. T. Smith. Remapped physical-virtual interfaces with bimanual haptic retargeting. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 19–27. IEEE, 2019. 2

[25] L. McAtamney and E. N. Corlett. Rula: a survey method for the investigation of work-related upper limb disorders. *Applied ergonomics*, 24(2):91–99, 1993. 3

[26] N. C. Nilsson, A. Zenner, and A. L. Simeone. Propping up virtual reality with haptic proxies. *IEEE Computer Graphics and Applications*, 41(5):104–112, 2021. 2

[27] A. Panarese and B. B. Edin. Human ability to discriminate direction of three-dimensional force stimuli applied to the finger pad. *Journal of neurophysiology*, 105(2):541–547, 2011. 3

[28] G. Robles-De-La-Torre and V. Hayward. Force can overcome object geometry in the perception of shape through active touch. *Nature*, 412(6845):445–448, 2001. 3

[29] S. S. Sawilowsky. New effect size rules of thumb. *Journal of modern applied statistical methods*, 8(2):26, 2009. 8

[30] H. Shi, H. Luo, H. Yi, and S. Je. Reachpad: Interacting with multiple virtual screens using a single physical pad through haptic retargeting. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pp. 1–17, 2025. 2

[31] A. Sun, J. Wu, R. Fan, S. K. Im, and L. Wang. Efficient and comfortable haptic retargeting with reset point optimization. *IEEE Transactions on Visualization and Computer Graphics*, 31(9):6588–6602, 2024. 3, 5

[32] X. Yang, Y. Kang, and X. Yang. Retargeting destinations of passive props for enhancing haptic feedback in virtual reality. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 618–619. IEEE, 2022. 3, 7, 9

[33] X. Yang, Y. Zhang, and X. Yang. Redirected placement: Evaluating the redirection of passive props during reach-to-place in virtual reality. In *Proceedings of the 29th ACM Symposium on Virtual Reality Software and Technology*, pp. 1–11, 2023. 5

[34] A. Zenner and A. Krüger. Estimating detection thresholds for desktop-scale hand redirection in virtual reality. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 47–55. IEEE, 2019. 3, 5

[35] A. Zenner, K. P. Regitz, and A. Krüger. Blink-suppressed hand redirection. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 75–84. IEEE, 2021. 3

[36] J. Zhang, W. Johal, and J. Knibbe. Illusion spaces in vr: The interplay between size and taper angle perception in grasping. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pp. 1–17, 2025. 3

[37] Y. Zhao and S. Follmer. A functional optimization based approach for continuous 3d retargeted touch of arbitrary, complex boundaries in haptic virtual reality. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, pp. 1–12, 2018. 2

[38] Y. Zhou and V. Popescu. Tapping with a handheld stick in vr: Redirec-

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.

tion detection thresholds for passive haptic feedback. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 83–92. IEEE, 2022. 3

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on November 10,2026 at 15:34:26 UTC from IEEE Xplore. Restrictions apply.